TryHackMe Writeup: What is Networking?

Field	Details
Name	Harini Edward
Intern ID	DG/15/OCT/CYBER/040
Program	DG Interns Hub, Week 1
Room	What is Networking?
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Estimated time	30 min
Status	Completed (100%)

Objective

Begin learning the fundamentals of computer networking: what a network is, how devices are identified on it and how to test that two devices can talk to each other.

What I did, task by task

What is the Internet?

- A network is a group of connected devices that can share data. The internet is a huge network made up of many smaller networks joined together.
- Looked at how the internet grew from small connected networks into the global network we use today.

Identifying devices: IP addresses

- An IP address is a logical address given to a device so other devices can find it on a network.
- An IPv4 address has four numbers (octets) separated by dots, each from 0 to 255, for example 192.168.1.10.
- Addresses can be public (used on the internet) or private (used inside a home or office network). IPv6 was introduced because IPv4 addresses are running out.

Identifying devices: MAC addresses

- A MAC address is a unique physical address set on the network card by the manufacturer. It has 12 hexadecimal characters, written in six pairs.
- The first half identifies the manufacturer and the second half identifies the device. A MAC address can be spoofed (faked), so it should not be trusted blindly.

Ping

- ping checks whether another device is reachable and how long it takes to respond. It uses the ICMP protocol.

- Your device sends an echo request and the other device sends back an echo reply. The time shown is the round trip time.

Commands and tools used

Command / Tool	What it does
ping	Test if a host is reachable and measure response time (ICMP)
ip a / ipconfig	Show the IP and MAC address of a device

Key takeaways

- A network connects devices, and the internet is a network of networks.
- IP addresses can change, while MAC addresses are fixed to the hardware (but can be spoofed).
- Public IPs are reachable on the internet and private IPs stay inside a local network.
- Ping is the quickest way to test connectivity. I will use it in the lab setup to check my VMs.

Why this matters for a SOC role

A SOC analyst reads IP and MAC addresses in almost every alert and log. Knowing what they mean and how to check connectivity is the base for network investigations.

Proof of completion

